Sang-Kyun Ko (고상균, 高像均)

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Google Scholar | LinkedIn | Personal Website | Academic Statement

RESEARCH PROFILE

Researcher working on machine learning for medical applications, with a focus on computer vision and flow-based generative modeling for coronary angiography and endoscopic imaging. Experienced in academic-hospital collaboration, applied research aimed at developing deployable products, and model development across image restoration, video frame interpolation, tracking, survival analysis, and cardiovascular risk prediction.

EMPLOYMENT

Machine Learning Researcher Oct 2023 – Present
Seoul National University Bundang Hospital (SNUBH), Coronary Artery AI Team, Cardiovascular Center, Internal Medicine
Research contract with professor

- Lead research on CAG image restoration from fluoroscopy to cine-quality sequences using flow- and diffusion-based models.
- Develop automated coronary artery tracking methods in CAG sequences for downstream analysis and clinical visualization support.
- Contributed to latent flow matching-based video frame interpolation for coronary angiography; related manuscript published in *npj Digital Medicine*.
- Led an atherosclerotic cardiovascular disease (ASCVD) risk prediction project using Kolmogorov-Arnold Networks (KANs) and health screening data.
- Investigate 3D reconstruction of coronary arteries from CAG sequences to improve anatomical interpretation and visualization.

Machine Learning Research Engineer Apr 2022 – Jul 2023
AINEX, Gastroscopic AI Part, AI Research and Development Team
Full-time employee

- Researched diffusion model-based endoscopic image generation for data augmentation and medical image analysis workflows.
- Developed gastroscopic image analysis models for diagnosis-support tasks, including real-time image classification.
- Worked on monocular object-size estimation for detected objects in endoscopic images.
- Collaborated with the AI R&D team on model experimentation, error analysis, and deployment-oriented validation.

Assistant Researcher Apr 2019 – Jul 2019
Industry-University Cooperation Foundation, Sun Moon University, System Software Research Lab.
Contract employee

- Developed CNN-based Chinese cursive character recognition methods for a university-industry research project.

PUBLICATIONS

1. **Ko, S.-K.**, Kim, S.-H., Kim, S.-H., Kim, H., Yoon, C.-H., Youn, T.-J., & Chae, I.-H. (2026). Kolmogorov-Arnold Network for Atherosclerotic Cardiovascular Disease Risk Prediction. *medRxiv*. [Under review]

2. Kwon, H., Park, S., Kim, D. H., **Ko, S.-K.**, Kim, S. H., Moon, I. T., et al. (2026). Video frame interpolation for coronary angiography using latent flow matching. *npj Digital Medicine*.

3. Moon, I. T.*, **Ko, S.-K.***, Kang, S. H., et al. (2024). Augmented Reality in Cardiology: Enhancing Visualization and Precision. *Current Cardiovascular Risk Reports*. *Equal contribution.

4. Jang, C., **Ko, S.**, Choi, J., Lim, J., Noh, Y. K., & Kim, T. J. (2022). Learning to increase matching efficiency in identifying additional b-jets in the $t\bar{t}b\bar{b}$ process. *The European Physical Journal Plus, 137*(7), 1–12.

CONFERENCES

1. **Ko, S.-K.**, Kang, S. H., Kim, S. H., Kim, H., Youn, T. J., Yoon, C. H., & Chae, I. H. (2026). Kolmogorov-Arnold Network for Atherosclerotic Cardiovascular Disease Risk Prediction. American College of Cardiology Scientific Session 2026 (ACC.26), Abstract Presentation. Published in *Journal of the American College of Cardiology, 87*(13 Suppl), A43. [Poster] [Project page]

2. **Ko, S.-K.** & Noh, Y.-K. (2023). Data Reconstruction Method for Learning to Censored Survival Data. *Korean Artificial Intelligence Association Summer Conference 2023*. Poster presentation. [Program book] [Paper] [Poster]

3. Oh, M. J., Park, J., Jeon, J., **Ko, S.-K.**, Jo, S., Kang, S., Kim, S. H., Park, S. H., Chang, Y. H., Shin, C. M., Kang, S. J., Kim, S. G., & Cho, S. J. (2023). Application of artificial intelligence in the detection of Borrmann type 4 advanced gastric cancer in upper endoscopy. *United European Gastroenterology Journal, 11*(Supplement 8), pp0253, p666. Poster presentation. [Abstract]

4. **Ko, S.**, Kim, B., & Kim, J. D. (2018). Deep Learning-based Algorithm for Object Identification in Multimedia. *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications* (*CUTE 2018*). Oral presentation. [Program book]

5. Park, K., Hong, B., **Ko, S.**, & Kim, B. (2018). Design and Implementation of Real-Time Web Authoring Tool Based on Drag-and-Drop Method. *The 13th KIPS International Conference on Ubiquitous Information Technologies and Applications* (*CUTE 2018*). Oral presentation. [Program book]

6. Seo, H., Hong, B., **Ko, S.-K.**, Park, K., Kim, B., & Jung, H. (2017). A survey on the performance of deep learning based on multiple GPUs. *Korea Software Congress 2017*. Poster presentation. [Program book] [Paper]

EDUCATION

Hanyang University (HYU), Seoul, Korea Sep 2019 – Feb 2022; Sep 2023 – Feb 2024
Advanced Application for Intelligence Systems Lab. (aAIs Lab), Department of Computer Science [Lab link]

- M.S. in Computer Science; advisor: Prof. Yung-Kyun Noh
- Thesis: "Learning to Directly Maximize Evaluation Metrics: Focus on Matching Efficiency and Concordance Index" [Thesis]
- Teaching Assistant: Artificial Intelligence (CSE4007), Automata Theory (ITE3061), Discrete Mathematics (MAT2020), Creative Software Programming: C++ (ITE1015)
- Led research on utilizing censored data as training data in survival analysis.
- Collaborated with particle physics researchers on identifying additional b-jets in the $t\bar{t}b\bar{b}$ process.

Sun Moon University, Asan, Korea Mar 2013 – Feb 2019
System Software Research Lab. (SSR Lab), Department of Computer Science
Later renamed Intelligent System and Software Research Lab., Chungbuk National University [Lab link]

- B.S. in Computer Science; advisor: Prof. Bongjae Kim
- Graduation project: "Deep Learning-based System for Object Retrieval in Multimedia"
- Student Excellence Scholarship, 2017

SHORT TERM EDUCATION

- Pattern Recognition and Machine Learning Winter School 2024, AI Society, Feb 2024 (12 hours, online). [Link]
- Pattern Recognition and Machine Learning Winter School 2021, AI Society, Feb 2021 (8 hours, online). [Link]
- The 11th KIAS CAC Online Winter School: Parallel Computing and Artificial Intelligence, Dec 2020 (20 hours, online). [Link]

AWARDS AND HONORS

- Nov 2018, **Excellence Prize**, 4th Industrial Revolution Talent Development Joint Festival, paper/coding division, Korea Society of Information Communication Security Ethics
- Nov 2018, **Grand Prize**, Comprehensive Competition for Computer Science and Electrical Engineering, Department of Computer Science, Sun Moon University
- Jun 2018, **Silver Prize**, Comprehensive Competition for Computer Science, Department of Computer Science, Sun Moon University
- Nov 2017, **Bronze Prize**, Global Capstone Design Workshop, Idea Factory, Sun Moon University
- Apr 2017, **Excellence Prize**, Idea Contest held by the President, Sun Moon University

CERTIFICATES

- Engineer Information Processing (정보처리기사), Human Resources Development Service of Korea, May 2019

MILITARY SERVICE

- Mar 2014 – Dec 2015, Republic of Korea Army, Infantry, Yanggu-gun, Gangwon-do; completed mandatory military service as a platoon signalman.

RESEARCH INTERESTS

- Flow-based generative modeling for medical imaging, video generation, and representation learning under distribution shift
- Addressing theoretical and statistical issues in machine learning for large-scale data analysis

REFERENCES

- **Yung-Kyun Noh** (nohyung@hanyang.ac.kr)
  Professor, Computer Science, Hanyang University
- **Bongjae Kim** (bjkim@chungbuk.ac.kr)
  Professor, Computer Science, Chungbuk National University